

CricMetrics Pro: Complete Machine Learning Models Guide

Easy-to-Understand Guide to Every Model Used in This Project (What, When, Why, How)

About this Guide: This document explains all the Machine Learning (ML) models used in the **CricMetrics Pro** cricket player performance system. It is written in simple, clear, and easy-to-understand English so that anyone—students, teachers, coaches, and team managers—can easily understand how each model works.

1. Quick Summary of All Models

In this project, we analyze **619 real IPL cricketers** across **17 seasons (2008–2024)** using **260,920 deliveries**. We use **6 Machine Learning models** plus **2 data preparation tools** across 3 main areas:

Area	Goal	Models Used	Best Model
1. Classification	Put players into 3 talent tiers: - Elite / Marquee (Tier 1) - Core / Star (Tier 2) - Developing / Squad (Tier 0)	- K-Nearest Neighbors (KNN) - Random Forest Classifier - Logistic Regression - Decision Tree	K-Nearest Neighbors (KNN) (95.16% Accuracy) & Random Forest (94.35%)
2. Clustering	Group players by tactical playing style without using preset labels	K-Means Clustering ($k=5$)	5 Tactical Archetypes (Anchor, Finisher, Fast Bowler, Spinner, All-Rounder)
3. Dimensionality Reduction	Compress 29 complex statistics into an easy 2D map (X, Y)	Principal Component Analysis (PCA)	2D Latent Map (Explains 60.8% of all variation)
4. Data Preparation	Clean and scale data so models can read it properly	- StandardScaler (Z-score scaling) - One-Hot Encoding	StandardScaler ($\mu=0, \sigma=1$)

2. Data Preparation Tools

Before feeding cricket numbers into machine learning models, we must prepare the data so computers can read it fairly.

2.1 StandardScaler (Z-Score Normalization)

- WHAT is it?**

A tool that rescales all numbers so that their average is 0 and standard deviation is 1. It turns big numbers and small numbers into the same standard scale:

$$\text{Scaled Value} = \frac{\text{Value} - \text{Average}}{\text{Spread (Std Dev)}}$$
- WHEN is it used?**

Immediately after splitting data into training (80%) and testing (20%). It is fitted only on training data and then used to scale test data and live user inputs in the web app.
- WHY do we need it?**

In cricket, `total_runs` can be huge (like 8,000 runs for Virat Kohli), while `economy_rate` is small (like 7.2 runs per over). If we don't scale them, distance-based models like KNN, K-Means, and PCA will pay 99% attention to runs and completely ignore bowling economy. Scaling makes every stat count fairly.

- **HOW is it implemented?**

We use `StandardScaler()` from Scikit-Learn. Saved to disk as `models/scaler.joblib`.

2.2 One-Hot Encoding

- **WHAT is it?**

A method that converts text categories (like `"All-Rounder"` or `"Specialist Bowler"`) into simple 0 and 1 columns that computers can calculate.

- **WHEN is it used?**

During the initial data cleaning step in `src/cricket_data_pipeline.py`.

- **WHY do we need it?**

Computers only understand numbers. But if we labeled roles as `1 = Batter, 2 = All-Rounder, 3 = Bowler`, the model would wrongly assume that a Bowler is "3 times bigger" than a Batter. One-hot encoding creates separate True/False (1/0) switches so no artificial rank is forced.

- **HOW is it implemented?**

Using Pandas `pd.get_dummies(df, columns=['primary_role'], drop_first=True)`. We drop the first column to avoid redundant mathematical loops (called the dummy variable trap).

3. Supervised Learning: Classification Models (Talent Tiers)

Classification models assign a player to one of **3 discrete talent categories**:

- **Tier 1 (Elite / Marquee)**: High-impact star match winners, captaincy candidates, high-value auction picks.
 - **Tier 2 (Core / Star)**: Consistent tournament starters and reliable anchors.
 - **Tier 0 (Developing / Squad)**: Emerging domestic prospects and squad backup options.
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3.1 K-Nearest Neighbors (KNN, $k=7$) — 🏆 TOP ACCURACY

- **WHAT is it?**

A model that classifies a player by finding the **7 most similar players** in cricket history and choosing the most common tier among those neighbors.

- **WHEN is it used?**

Used in Tab 2 and Tab 6 for talent tier classification.

- **WHY do we need it?**

1. **Highest Accuracy**: Achieved **95.16% test accuracy** (top among all classifiers).

2. **Matches Real Cricket Scouting**: When a franchise scout looks at an uncapped player, they naturally ask: *"Who does this player resemble?"* If a young player matches Hardik Pandya, Ben Stokes, and Andre Russell in batting impact, strike rate, and overs bowled, they naturally belong in Tier 1.

- **HOW does it work & performance?**

- Calculates Euclidean distance across all 29 scaled stats:

$$\text{Distance} = \sqrt{\sum (x_i - y_i)^2}$$

- Finds the 7 closest historical players and takes a distance-weighted vote.

- **Test Accuracy:** 95.16% | **Macro F1-Score:** 0.9501 | **Precision:** 0.9601 | **Recall:** 0.9421 .
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3.2 Random Forest Classifier — 🏆 PRODUCTION TIER ENGINE

- **WHAT is it?**

A team of **150 decision trees** voting together on which tier a player belongs to.

- **WHEN is it used?**

Powers the Tier Classification engine and the **Feature Importance Leaderboard** in Tab 6.

- **WHY do we need it?**

1. **High & Stable Accuracy:** Achieved **94.35% test accuracy** and the highest cross-validation score (**94.75% across 5 folds**).

2. **Feature Importance (Explainability):** It tells franchise owners which stats matter most using Gini impurity.

- **HOW does it work & performance?**

- Built using `RandomForestClassifier(n_estimators=150, max_depth=10, random_state=42)` .

- **Test Accuracy:** 94.35% | **5-Fold CV Score:** 94.75% | **Macro F1:** 0.9459 .

- **Top Factors Identified:** Clutch Match-Winner Index (12.8%), Batting Impact Index (11.6%), Player of the Match awards (7.8%).

3.3 Decision Tree Classifier (CART)

- **WHAT is it?**

A visual flowchart of simple "IF-THEN" rules:

- *Question 1:* Is `clutch_match_winner_index` > 24.5?
- *If Yes:* Is `batting_impact_index` > 62.0? $\rightarrow$ **Elite Tier**
- *If No:* Is `wickets_taken` < 8? $\rightarrow$ **Developing Tier**

- **WHEN is it used?**

Used when coaches or non-technical executives want a clear, rule-by-rule explanation without complicated math.

- **WHY do we need it?**

It is the most interpretable model in data science. Anyone can trace the decision path on a sheet of paper.

- **HOW does it work & performance?**

- Uses the Gini Impurity formula to pick questions that cleanly split players into tiers.

- **Test Accuracy:** 87.90% | **Macro F1-Score:** 0.8812 .

3.4 Logistic Regression (Multinomial / Softmax)

- **WHAT is it?**

A linear classification model that gives the exact probability percentage for each tier (for example: 85% Elite, 12% Core, 3% Developing).

- **WHEN is it used?**

Used as a baseline probabilistic classifier.

- **WHY do we need it?**

Franchise directors like to see risk percentages during auction bidding wars rather than just a flat "Yes/No" label.

- **HOW does it work & performance?**

- Computes linear scores for each tier and uses the Softmax mathematical function to turn scores into probabilities that sum to 100%.
 - **Test Accuracy:** 87.90% | **5-Fold CV Score:** 92.53% | **Macro F1-Score:** 0.8853 .
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4. Unsupervised Learning: Clustering (Tactical Archetypes)

Unlike classification, clustering has **no correct answer labels**. The computer looks at all players and groups them by similar playing styles on its own.

4.1 K-Means Clustering ($k=5$)

- **WHAT is it?**

An algorithm that groups 619 cricketers into **5 tactical clusters** based on how close their stats are in multi-dimensional space.

- **WHEN is it used?**

Powers **Tab 4 (Tactical Archetypes & Roster Balance)** in the web app.

- **WHY do we need it?**

Nominal labels like "Batter" or "Bowler" are too simple for modern T20 cricket:

- Virat Kohli and Andre Russell are both listed as "Batters", but they play completely different roles (Kohli anchors the innings; Russell hits explosive boundaries in death overs).
 - Jasprit Bumrah and Yuzvendra Chahal are both "Bowlers", but Bumrah bowls 145 km/h yorkers at the death while Chahal spins middle-over balls.
 - K-Means automatically discovered these **5 real tactical archetypes**:
 1. **Cluster 0: Tactical Anchor & Top-Order Accumulator** (builds partnerships, high average).
 2. **Cluster 1: High-Impact Pace Spearhead & Death Bowler** (yorkers, low death economy, high wickets).
 3. **Cluster 2: Explosive Death-Over Finisher & Boundary Hitter** (strike rate > 155, high sixes).
 4. **Cluster 3: Mystery / Control Spin Maestro** (high dot ball %, economical middle overs).
 5. **Cluster 4: Elite Dual-Threat All-Rounder** (contributes heavily with both bat and ball).
 - **HOW does it work?**
 1. We tested cluster counts from $k=2$ to $k=7$ using the **Elbow Curve** (measuring cluster tightness/inertia) and **Silhouette Score** (measuring how distinct clusters are from each other).
 2. $k=5$ gave the clearest real-world cricket separation ($s = 0.285$).
 3. Players are assigned to the nearest cluster center (centroid).
 - Saved as `models/kmeans_model.joblib`.
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5. Dimensionality Reduction (Visualizing High-Dimensional Data)

5.1 Principal Component Analysis (PCA, 2 Components)

- **WHAT is it?**

A mathematical technique that compresses 29 different statistics down into **2 summary numbers (Component 1 and Component 2)** so that all 619 players can be drawn on a standard 2D scatter plot (X and Y axis).

- **WHEN is it used?**

Powers the interactive 2D Tactical Scatter Plot in **Tab 4** of the web app.

- **WHY do we need it?**

Human eyes and computer screens can only view 2 or 3 dimensions at once. Looking at a 29-column spreadsheet is overwhelming. PCA keeps the most important information while allowing coaches to see every player's tactical position on a single screen.

- **HOW does it work & performance?**

- It finds the two directions where player statistics vary the most (using eigenvalues of the covariance matrix):
- **Principal Component 1 (X-axis, 38.16% variance):** Represents general match volume and career impact (runs, wickets, matches).
- **Principal Component 2 (Y-axis, 22.63% variance):** Represents bowling vs. batting specialization (bowlers go up, batters go down, all-rounders stay in the middle).
- **Together, these 2 numbers capture 60.79% of all information** in the 29 original columns.
- Saved as `models/pca_model.joblib`.

6. Model Evaluation & Validation Techniques

To ensure our models actually work on unseen matches and don't just memorize the past, we use rigorous testing methods:

1. 80/20 Stratified Train/Test Split (Module VI):

- 495 players (80%) are used to train the models.
- 124 players (20%) are locked away in a vault and only used to test final performance.
- "Stratified" means both sets have the exact same percentage of Elite, Core, and Developing players.

2. 5-Fold Stratified Cross-Validation (Module VI):

- The training set is split into 5 equal parts. The model trains on 4 parts and tests on the 5th part, repeating this 5 times.
- We report the average and standard deviation (e.g. 0.9475 ± 0.0142). This proves the model is reliable and didn't just get lucky.

3. Evaluation Metrics Used:

- **Accuracy:** Percentage of correct tier predictions (KNN got 95.76% , Random Forest got 94.35%).
 - **Macro F1-Score:** Balances precision and recall across small and large tiers so minority classes aren't ignored.
 - **Precision & Recall:** Evaluates exact class sensitivity.
 - **Confusion Matrix:** A 3×3 grid showing where the model was right and where it made mistakes.
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7. Master Model Leaderboard & Comparison

Model Name	Task	Syllabus Module	Test Score	5-Fold CV Score	Why We Chose It
K-Nearest Neighbors (KNN)	Tier Classification	Module V	95.16\% Acc (F1: 0.9501)	0.9030 \pm 0.0221	Accuracy Winner: Classifies players by finding historical peer matches.
Random Forest Classifier	Tier Classification	Module VIII	94.35\% Acc (F1: 0.9459)	0.9475 \pm 0.0142	Production Winner: Stable across folds and gives Top 10 feature rankings.
Logistic Regression	Tier Classification	Module V	87.90\% Acc (F1: 0.8853)	0.9253 \pm 0.0180	Linear probabilistic model; gives exact risk percentages.
Decision Tree (CART)	Tier Classification	Module V	87.90\% Acc (F1: 0.8812)	0.8990 \pm 0.0195	100% human-readable "IF-THEN" flowchart for coaches.
K-Means Clustering	Tactical Styles	Module VII	5 Clusters ($s = 0.285$)	Validated via Elbow Curve	Discovered 5 real playing styles beyond simplistic Batter/Bowler tags.
PCA	2D Visualization	Module VIII	60.79\% Var (2 Components)	SVD Decomposition	Compresses 29 stats into an interactive 2D scatter plot.

Key Takeaway for Viva & Interviews

When asked: "Why did you use these specific models?"

Answer:

"Every model was chosen to fulfill our university syllabus and solve a real sports business problem. For talent tiers, **K-Nearest Neighbors** achieved top test accuracy (95.16\%) because evaluating players against similar historical peers is the natural way sports scouting works. **Random Forest Classifier** provides enterprise stability (94.75\% 5-fold CV) and Gini feature rankings. **Logistic Regression** outputs exact class probabilities, and **Decision Tree** gives a clear visual flowchart. Finally, **K-Means (k=5)** and **PCA** allow coaches to discover tactical playing styles and visualize complex 29-dimensional performance on a simple 2D map."